

University of Dhaka
Affiliated Engineering Colleges
Department of Computer Science and Engineering
1st Year 1st Semester B.Sc. Examination, 2020
MATH – 1105, Differential and Integral Calculus

Total Marks: 70

Time: 2 Hours

(Answer any 3 (Three) of the following Questions)

1. a) Define continuity. A function $f(x)$ is defined as follows: 7.33

$$f(x) = \begin{cases} 3 + 2x & \text{for } -3/2 \leq x < 0 \\ 3 - 2x & \text{for } 0 \leq x < 3/2 \\ -3 - 3x & \text{for } x \geq 3/2 \end{cases}$$
show that $f(x)$ is continuous at $x = 0$.
- b) Draw the graphs of the following function. If $y = \begin{cases} 1 & \text{when } x > 0 \\ 0 & \text{when } x = 0 \\ -1 & \text{when } x < 0 \end{cases}$ 8
- c) Do the following limit exist $\lim_{x \rightarrow \frac{\pi}{2}} \left(\frac{e^{\tan x} - 1}{e^{\tan x} + 1} \right)$ 8
2. a) Show that the maximum value of $\left(\frac{1}{x}\right)^x$ is $e^{\frac{1}{e}}$ 7
- b) State L'Hospital rule. Using L'Hospital rule to evaluate the following limits: 2.33
+7
i) $\lim_{\theta \rightarrow \frac{\pi}{4}} \frac{\sqrt{2} - \cos \theta - \sin \theta}{(4\theta - \pi)^2}$ ii) $\lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}}$
- c) Show that the maximum value of $x + \frac{1}{x}$ is less than its minimum value. 7
3. a) What is Taylor's infinite series? Using Taylor's theorem prove that 3+
 $\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5(1+c)^5}$, where $0 < c < x$ 5.33
- b) Differentiate $\tan^{-1} \frac{2x}{1-x^2}$ with respect to $\sin^{-1} \frac{2x}{1+x^2}$ 7
- c) Find the equation of the tangent and normal at the point (x, y) of the curve $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$ 8
4. a) Find the area above the x-axis, included between the parabola $x^2 = ax$ and the circle $x^2 + y^2 = 2ax$ 8
- b) Find the area of the quadrant of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ between the major and minor axes. 8
- c) show that $\int_0^{\pi} x \log \sin x dx = \frac{\pi^2}{2} \log \frac{1}{2}$ 7.33
5. a) Evaluate the followings (any two): 10
(i) $\int_0^{\frac{\pi}{2}} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx$ (ii) $\int_0^{\frac{\pi}{2}} \frac{dx}{1 + \cot x}$ (iii) $\int_0^1 x^3 \sqrt{1+3x^4} dx$
- b) Evaluate the followings (any two): 8
(i) $\int \frac{x + \sin x}{1 + \cos x} dx$ (ii) $\int \frac{dx}{(\sin x + \cos x)^2} dx$ (iii) $\int \frac{dx}{a + b \sin x}$
- c) Show that $\lim_{x \rightarrow 0} \left(\frac{x - \sin x}{x^3} \right) = 1/6$ 5.33

1. (a)

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□ Continuity of a function:

A function $f(x)$ is said to be continuous at $x=a$ if it exists and equal to $f(a)$.
In other words, $f(x)$ is said to be a continuous function at $x=a$ if,

$$\lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow a^-} f(x) = f(a)$$

$$1.(b) f(x) = \begin{cases} 3+2x & -\frac{3}{2} \leq x < 0 \\ 3-2x & \text{when } 0 \leq x < 3/2 \\ -3-2x & x \geq 3/2 \end{cases}$$

For $x=0$,

when $x > 0$ then $f(x) = 3-2x$

$$\therefore \text{R.H.L} = \lim_{x \rightarrow 0^+} f(x)$$

$$= \lim_{h \rightarrow 0} f\left(\frac{0}{2} + h\right)$$

$$= \lim_{h \rightarrow 0} (0 + h)$$

$$= \lim_{h \rightarrow 0} \{3 - 2(h)\}$$

$$= \{3 - 2(0)\}$$

$$= 3$$

When, $x < 0$ then $f(x) = 3 + 2x$

$$\therefore \text{L.H.L} = \lim_{x \rightarrow 0^-} f(x)$$

$$= \lim_{h \rightarrow 0} f(0-h)$$

$$= \lim_{h \rightarrow 0} \{3 + 2(h)\}$$

$$= \{3 + 0\}$$

$$= 3$$

$\therefore$ Functional value, when $x = 0$

$$\text{F.V} = f(x) = 3 - 2x$$

$$\Rightarrow f(0) = 3 - 2(0) \\ = 3 - 0$$

$$\therefore f(0) = 3$$

$$\therefore \text{R.H.L} = \text{L.H.L} = \text{F.V}$$

$\therefore$ The function $f(x)$ is continuous at $x = 0$.

① ③

given that,

$$\lim_{u \rightarrow \frac{\pi}{2}} \frac{e^{\tan u} - 1}{e^{\tan u} + 1}$$

Now,

$$\text{right hand limit} = \lim_{u \rightarrow \frac{\pi}{2}^+} \frac{e^{\tan u} - 1}{e^{\tan u} + 1}$$

$$\begin{aligned} &= \frac{e^{-\infty} - 1}{e^{-\infty} + 1} \quad \left| \begin{array}{l} \tan u \rightarrow -\infty \\ [u = \frac{\pi}{2}^+] \end{array} \right. \\ &= \frac{0 - 1}{0 + 1} \\ &= -1 \end{aligned}$$

$$\begin{aligned} \text{left hand limit} &= \lim_{u \rightarrow \frac{\pi}{2}^-} \frac{e^{\tan u} - 1}{e^{\tan u} + 1} \\ &= \lim_{u \rightarrow \frac{\pi}{2}^-} \frac{1 - e^{\frac{1}{\tan u}}}{1 + e^{\frac{1}{\tan u}}} \quad \left| \begin{array}{l} \therefore \tan u \rightarrow +\infty \\ [u = \frac{\pi}{2}^-] \end{array} \right. \\ &= \frac{1 - 0}{1 + 0} \end{aligned}$$

$\neq 1$; since right hand limit $\neq$ left hand limit

the given limit does not exist.

(A)

2.a

$$\text{Let } y = \left(\frac{1}{x}\right)^x$$

$$\Rightarrow \ln y = x \ln \frac{1}{x}$$

Differentiating on both sides gives,

$$\Rightarrow \frac{1}{y} \cdot y' = \ln \frac{1}{x} + \frac{x}{1} \cdot \left(\frac{-1}{x^2}\right)$$

$$\Rightarrow \frac{y'}{y} = \ln \frac{1}{x} - 1$$

$$\Rightarrow y' = \left(\frac{1}{x}\right)^x \left(\ln \frac{1}{x} - 1\right)$$

At maximum value, $y' = 0, y'' < 0$

$$y' = 0 \Rightarrow \ln \frac{1}{x} = 1 \Rightarrow \frac{1}{x} = e \Rightarrow x = \frac{1}{e}$$

$$y'' = \left(\frac{1}{x}\right)^x (\ln x - 1)^2 + \left(\frac{1}{x}\right)^x \cdot \frac{-1}{x}$$

$$y'' \left(\frac{-1}{e}\right) < 0.$$

$\therefore$ At $x = \frac{1}{e}$, y has maximum value equal to $(e)^{\frac{1}{e}}$

2. (b) L'Hospital's Rule:

statement: If $\phi(x)$, $\psi(x)$ as also their derivatives.

$\phi(x)$ and $\psi(x)$ are continuous at $x=a$ and if $\phi(a) = \psi(a) = 0$.

[i.e; $\lim_{x \rightarrow a} \phi(x) = \lim_{x \rightarrow a} \psi(x) = 0$] then,

$$\lim_{x \rightarrow a} \frac{\phi(x)}{\psi(x)} = \lim_{x \rightarrow a} \frac{\phi'(x)}{\psi'(x)} = \frac{\phi'(a)}{\psi'(a)} = \psi'(a) \neq 0$$

Example: Evaluate $\lim_{x \rightarrow 0} \frac{e^x - e^{-x} - 2x}{x - \sin x}$ by

L'Hospital's rule.

Solve: Given,

$$\lim_{x \rightarrow 0} \frac{e^x - e^{-x} - 2x}{x - \sin x} \left[\frac{0}{0} \text{ form} \right]$$

$$= \lim_{x \rightarrow 0} \frac{e^x + e^{-x} - 2}{1 - \cos x} \left[\frac{0}{0} \text{ form} \right]$$

$$= \lim_{x \rightarrow 0} \frac{e^x - e^{-x} - 0}{+ \sin x} \left[\frac{0}{0} \text{ form} \right]$$

$$= \lim_{x \rightarrow 0} \frac{e^x + e^{-x}}{\cos x}$$

$$= \frac{e^0 + e^{-0}}{\cos 0} = \frac{1+1}{1} = 2 \quad (\text{Ans})$$

$$\textcircled{i} \lim_{\theta \rightarrow \frac{\pi}{4}} \frac{\sqrt{2} - \cos \theta - \sin \theta}{(4\theta - \pi)^2} \left[\frac{0}{0} \text{ form} \right]$$

$$= \lim_{\theta \rightarrow \frac{\pi}{4}} \frac{0 + \sin \theta - \cos \theta}{2(4\theta - \pi) \times 4} \left[\frac{0}{0} \text{ form} \right]$$

$$= \lim_{\theta \rightarrow \frac{\pi}{4}} \frac{\cos \theta + \sin \theta}{8 \times 4}$$

$$= \frac{\cos \frac{\pi}{4} + \sin \frac{\pi}{4}}{32}$$

$$= \frac{\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}}{32}$$

$$= \frac{\frac{2}{\sqrt{2}}}{32} = \frac{\sqrt{2}}{32} \quad (\text{Ans})$$

$$ii) \lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}}$$

Let,

$$L = \lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}}$$

$$\Rightarrow \ln L = \lim_{x \rightarrow 0} \ln (1+x)^{\frac{1}{x}}$$

$$\Rightarrow \ln L = \lim_{x \rightarrow 0} \frac{1}{x} \ln (1+x)$$

$$\Rightarrow \ln L = \lim_{x \rightarrow 0} \frac{\ln (1+x)}{x} \left[\frac{0}{0} \text{ form} \right]$$

$$\Rightarrow \ln L = \lim_{x \rightarrow 0} \frac{\frac{1}{1+x}}{1}$$

$$\Rightarrow \ln L = \lim_{x \rightarrow 0} \frac{1}{1+x}$$

$$\Rightarrow \ln L = \frac{1}{1} = 1$$

$$\Rightarrow \ln L = 1$$

$$\therefore L = e^1 = e$$

$$\therefore \lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}} = e$$

(Ans)

c)

Let,

$$f(x) = x + \frac{1}{x}$$

$$f'(x) = 1 - \frac{1}{x^2} \quad \text{--- (i)}$$

For a maximum or a minimum values of $f(x)$, $f'(x) = 0$.

i.e;

$$1 - \frac{1}{x^2} = 0$$

$$\frac{1}{x^2} = 1$$

$$\Rightarrow x^2 = 1$$

$$\therefore x = \pm 1$$

$$\text{Again, } f''(x) = 0 + \frac{2}{x^3}$$

$$= \frac{2}{x^3}$$

when, $x = +1$, then,

$$f''(1) = \frac{2}{1} = 2 \text{ (+ve)}$$

$\therefore f(x)$ is ~~min~~ minimum when, $x = +1$ and the minimum values is $f(1) = 1 + \frac{1}{1} = 2$

when, $x = -1$, then,

$$f''(-1) = \frac{2}{(-1)^3} = -2 \quad (-ve)$$

$f(x)$ is maximum when $x = -1$ and
the maximum value is $f(x) = -2 + \frac{1}{(-2)}$

$$0 = (x)' \quad (x) = -2 + \frac{1}{2}$$

$$0 = \frac{1}{x} - 1 \quad = \frac{-4-1}{2}$$

$$1 = \frac{1}{x}$$

$$= \frac{-5}{2}$$

$$1 = \frac{1}{x} \Rightarrow$$

$$= -2.5$$

$\therefore$ max value $<$ minimum value.

$\therefore$ maximum value of $x + \frac{1}{x}$ is less
than its minimum value (showed)

3.(a)

Taylor's infinite series ~~also known as~~ is a mathematical representation of a function as an infinite sum of terms, each of which is derivative of the function evaluated at a particular point.

The general form of Taylor's series for a function $f(x)$ centered at $x = a$ is:

$$f(x) = f(a) + f'(a)(x-a)/1! + f''(a)(x-a)^2/2! + f'''(a)(x-a)^3/3! + \dots$$

$$\ln(1+x) = \int_0^x \frac{1}{1+t} dt.$$

Then we integrate the right-hand side of (1) term by term. We get

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots,$$

precisely the same thing as what one gets by putting $a = 0$ in your expression.

$$b) \frac{d}{d\left(\sin^{-1} \frac{2x}{1+x^2}\right)} \tan^{-1}\left(\frac{2x}{1-x^2}\right)$$

$$= \frac{d}{d(\tan^{-1} x)} 2 \tan^{-1} x$$

$$= 1$$

we know,

$$\tan^{-1} \frac{2x}{1-x^2} = 2 \tan^{-1} x$$

$$\sin^{-1} \frac{2x}{1+x^2} = 2 \tan^{-1} x$$

c) Here, $x^{2/3} + y^{2/3} = a^{2/3}$

$$\frac{2}{3} x^{2/3-1} + \frac{2}{3} y^{2/3-1} \frac{dy}{dx} = 0$$

$$\Rightarrow x^{-1/3} + y^{-1/3} \frac{dy}{dx} = 0$$

$$\Rightarrow y^{-1/3} \frac{dy}{dx} = -x^{-1/3}$$

$$\Rightarrow \frac{dy}{dx} = -x^{-1/3} \cdot y^{1/3} \quad \left[y^{-1/3} = \frac{1}{y^{1/3}} \right]$$

$$\Rightarrow \frac{dy}{dx} = -\left(\frac{y}{x}\right)^{1/3} \quad \left[x^{-1/3} = \frac{1}{x^{1/3}} \right]$$

$$\text{at } (x, y) \quad \frac{dy}{dx} = -\left(\frac{y}{x}\right)^{1/3}$$

So at (x, y) point equation of tangent is,

$$y_1 - y_0 = \left(\frac{dy}{dx} \right)_{(x_0, y_0)} (x_1 - x_0)$$

$$\Rightarrow \frac{dy}{dx} (x_0, y_0) = \frac{y_1 - y_0}{x_1 - x_0}$$

$$\Rightarrow \frac{dy}{dx} (x, y) = \frac{y_1 - y}{x_1 - x}$$

$$\Rightarrow -\left(\frac{y}{x}\right)^{1/3} = \frac{y_1 - y}{x_1 - x}$$

$$\Rightarrow -y^{1/m}(x_1-x) = x^{1/m}(y_1-y)$$

$$\Rightarrow -y^{1/m}(x_1-x) - x^{1/m}(y_1-y) = 0$$

$$\therefore x^{1/m}(y_1-y) + y^{1/m}(x_1-x) = 0$$

$$(x-h)^2 + (y-k)^2 = r^2$$

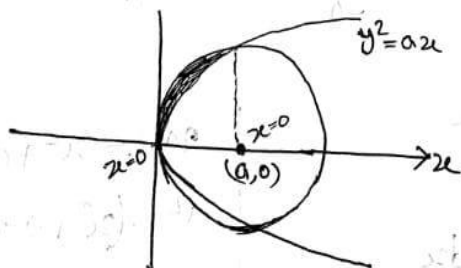
$$x^2 + y^2 = 2ax$$

$$x^2 + y^2 - 2ax = 0$$

$$x^2 - 2ax + y^2 + a^2 = a^2$$

$$(x-a)^2 + (y-0)^2 = a^2$$

Centre = $(a, 0)$ $r = a$



$$x^2 + y^2 - 2ax = 0$$

$$\Rightarrow x^2 + ax - 2ax = 0$$

$$x^2 - ax = 0$$

$$x(x-a) = 0$$

$$x=0 ; x=a$$

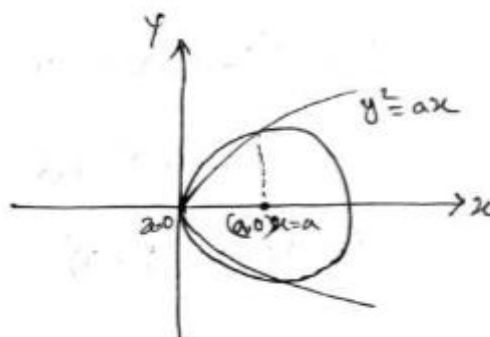
$$y^2 = ax$$

$$y^2 = a(a)$$

$$y = \pm a$$

4. (a)

Ans: The Points of intersection of both the curves are $(0,0)$ and (a,a)



$$\begin{aligned} \text{Area} &= \int_0^a (\sqrt{2ax-x^2} - \sqrt{ax}) dx \\ &= \int_0^a \sqrt{2ax-x^2} dx - \int_0^a \sqrt{ax} dx \quad \text{--- (1)} \end{aligned}$$

$$\begin{aligned} I_1 &= \int_0^a \sqrt{2ax-x^2} dx \\ &= \int_0^a \sqrt{a^2 - (x-a)^2} dx \end{aligned} \quad \left| \begin{array}{l} 2ax - x^2 + a^2 - a^2 \\ a^2 - (x^2 + a^2 - 2ax) \\ a^2 - (x-a)^2 \end{array} \right.$$

Put $x-a = a \sin \theta$
 $dx = a \cos \theta \cdot d\theta$

When $x=0$ $-a = a \sin \theta$
 $\Rightarrow -1 = \sin \theta$
 $\sin^{-1}(-1) = \theta$
 $-\frac{\pi}{2} = \theta$

When $x=a$,
 $0 = a \sin \theta$
 $\sin \theta = 0$
 $\theta = 0$

Now,

$$I_1 = \int_{-\pi/2}^0 \sqrt{a^2 - (a \sin \theta)^2} (a^2 \cdot \cos \theta \cdot d\theta)$$

$$= a^2 \int_{-\pi/2}^0 \cos^2 \theta \, d\theta$$

$$= a^2 \int_{-\pi/2}^0 \frac{\cos 2\theta + 1}{2} d\theta$$

$$= \frac{a^2}{2} \left[\int_{-\pi/2}^0 \cos 2\theta \, d\theta \cdot \int_{-\pi/2}^0 1 \, d\theta \right]$$

$$= \frac{a^2}{2} \left[\sin 2\theta + \theta \right]_{-\pi/2}^0$$

$$= \frac{a^2}{2} \left[0 + 0 - \sin(-\pi) - \left(-\frac{\pi}{2}\right) \right]$$

$$= \frac{a^2}{2} \left[\frac{\pi}{2} \right]$$

$$= \frac{a^2 \pi}{4}$$

$$I_2 = \int_0^a \sqrt{ax} \, dx$$

$$= \sqrt{a} \int_0^a x^{\frac{1}{2}} \, dx$$

$$= \sqrt{a} \left[\frac{x^{\frac{1}{2}+1}}{\frac{1}{2}+1} \right]_0^a$$

$$= \sqrt{a} \cdot \frac{2}{3} [a^{\frac{3}{2}} - 0]$$

$$= \frac{2}{3} a^{\frac{3}{2}}$$

$$\text{Area} = I_1 - I_2 \left[0 + 0 \sin^2 \right]$$

$$= \frac{a^2 \pi}{4} - \frac{2a^2}{3}$$

$$= a^2 \left(\frac{\pi}{4} - \frac{2}{3} \right)$$

(Ans)

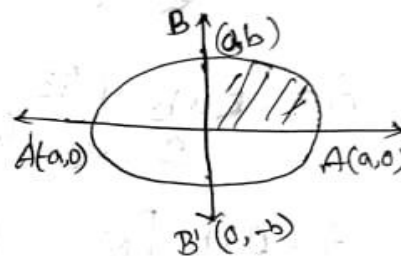
4. (b)

ଚିତ୍ରରେ ଦିଆଯାଇଛି $y=0$

$$\frac{x^2}{a^2} = 1 \quad x = \pm a$$

$$x=0$$

$$\Rightarrow \frac{y^2}{b^2} = 1 \quad y = \pm b$$



Ans Area of the ellipse

~~=~~ Area of OAB

$$= \int_0^a \frac{b}{a} \sqrt{a^2 - x^2} dx$$

$$= \frac{b}{a} \int_0^a \sqrt{a^2 - x^2} dx$$

$$x = a \sin \theta$$

$$dx = a \cos \theta \cdot d\theta$$

Lower limit $\theta = 0$ $\theta = 0$

Upper limit $a = a \sin \theta$
 $\theta = \frac{\pi}{2}$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\frac{y^2}{b^2} = 1 - \frac{x^2}{a^2}$$

$$y = \pm \frac{b}{a} \sqrt{a^2 - x^2}$$

$$A = \cancel{\frac{b}{a}} \int_0^{\pi/2} \sqrt{a^2 - a^2 \sin^2 \theta} \cdot a \cos \theta \cdot d\theta$$

$$A = \cancel{\frac{b}{a}} \times a^2 \int_0^{\pi/2} \cos \theta \cdot \cos \theta \cdot d\theta$$

$$A = \frac{ab}{2} \int_0^{\pi/2} 2 \cos^2 \theta \cdot d\theta$$

$$A = \frac{ab}{2} \int_0^{\pi/2} (1 + \cos 2\theta) d\theta$$

$$A = \frac{ab}{2} \left[\theta + \frac{\sin 2\theta}{2} \right]_0^{\pi/2}$$

$$A = \frac{ab}{2} \left[\left(\frac{\pi}{2} + 0 \right) - (0 + 0) \right]$$

$$A = \frac{ab}{2} \times \frac{\pi}{2}$$

$$A = \frac{\pi ab}{4} \text{ Ans}$$

4. C)
Solution



Correct option is D)

$$I = \int_0^{\pi} x \log(\sin x) dx \dots (i)$$

$$I = \int_0^{\pi} (\pi - x) \log(\sin(\pi - x)) dx$$

$$I = \int_0^{\pi} (\pi - x) \log(\sin x) dx \dots (ii)$$

Adding (i) and (ii), we get

$$2I = \pi \int_0^{\pi} \log(\sin x) dx$$

$$2I = 2\pi \int_0^{\frac{\pi}{2}} \log(\sin x) dx$$

$$\Rightarrow I = \pi \left(-\frac{\pi}{2} \log 2 \right)$$

$$\Rightarrow I = -\frac{\pi^2}{2} \log(2) = \frac{\pi^2}{2} \log\left(\frac{1}{2}\right)$$

5. a) i

Integral calculus:

$$1. \text{ Let } I = \int_0^{\pi/2} \frac{\sqrt{\sin u}}{\sqrt{\sin u} + \sqrt{\cos u}} \quad \text{--- (i)}$$

$$\int_0^a f(u) du = \int_0^a f(a-u) du$$

$$= \int_0^{\pi/2} \frac{\sqrt{\sin(\pi/2 - u)}}{\sqrt{\sin(\pi/2 - u)} + \sqrt{\cos(\pi/2 - u)}} (du)$$

$$I = \int_0^{\pi/2} \frac{\sqrt{\cos u}}{\sqrt{\cos u} + \sqrt{\sin u}} du \quad \text{--- (ii)}$$

Now (i) + (ii) $\Rightarrow$

$$I + I = \int_0^{\pi/2} \frac{\sqrt{\sin u}}{\sqrt{\sin u} + \sqrt{\cos u}} du + \int_0^{\pi/2} \frac{\sqrt{\cos u}}{\sqrt{\sin u} + \sqrt{\cos u}} du$$

$$= \int_0^{\pi/2} \left(\frac{\sqrt{\sin u} + \sqrt{\cos u}}{\sqrt{\sin u} + \sqrt{\cos u}} \right) du$$

$$= \left[u \right]_0^{\pi/2}$$

$$2I = \pi/2 - 0$$

$$\Rightarrow I = \pi/4$$

② $\pi/2$

$$\int_0^{\pi/2} \frac{du}{1 + \cot u}$$

$$\text{Let } I = \int_0^{\pi/2} \frac{1}{1 + \cot u} du = \int_0^{\pi/2} \frac{1}{1 + \frac{\cos u}{\sin u}} du$$

$$I = \int_0^{\pi/2} \frac{1}{\frac{\sin u + \cos u}{\sin u}} du = \int_0^{\pi/2} \frac{\sin u}{\sin u + \cos u} du \quad \text{--- (i)}$$

$$I = \int_0^{\pi/2} \frac{\sin(\pi/2 - u)}{\sin(\pi/2 - u) + \cos(\pi/2 - u)} du \quad \left(\int_0^a f(u) du = \int_0^a f(a-u) du \right)$$

$$I = \int_0^{\pi/2} \frac{\cos u}{\cos u + \sin u} du \quad \text{--- (ii)}$$

Adding (i) and (ii), we get

$$2I = \int_0^{\pi/2} \left(\frac{\sin u}{\sin u + \cos u} + \frac{\cos u}{\sin u + \cos u} \right) du$$

$$2I = \int_0^{\pi/2} \left(\frac{\cancel{\sin u + \cos u}}{\cancel{\sin u + \cos u}} \right) du$$

$$2I = \int_0^{\pi/2} du = [u]_0^{\pi/2} = \pi/2 - 0$$

$$2I = \pi/2$$

$$\therefore I = \pi/4$$

5. d) ii

$$\int_0^{\pi/2} \frac{1 \, d\alpha}{1 + \cot \alpha}$$

$$= \int_0^{\pi/2} \frac{1 \, du}{1 + \frac{\cos u}{\sin u}}$$

$$= \int_0^{\pi/2} \frac{\sin u}{\sin u + \cos u}$$

$$= \frac{1}{2} \int_0^{\pi/2} \frac{\sin u + \cos u - \cos u + \sin u}{\sin u + \cos u}$$

$$= \frac{1}{2} \int_0^{\pi/2} \left(\frac{\sin u + \cos u}{\sin u + \cos u} + \frac{\sin u - \cos u}{\cos u + \sin u} \right) du$$

$$= \frac{1}{2} \int_0^{\pi/2} du - \frac{1}{2} \int_0^{\pi/2} \frac{-\sin u + \cos u}{\cos u + \sin u} du$$

$$= \frac{1}{2} \int_0^{\pi/2} \frac{1}{2} \left[u \right]_0^{\pi/2} -$$

$$\frac{1}{2} \int_0^1 \frac{1}{2} dz$$

$$\begin{aligned} z &= \cos u + \sin u \\ dz &= (-\sin u + \cos u) du \end{aligned}$$

u	$\pi/2$	0
z	1	1

$$= \frac{1}{2} \left[x \right]_0^{\pi/2} - \frac{1}{2} \left[\ln x \right]_1^1$$

$$= \frac{\pi}{4} - \frac{1}{2} (\ln 1 - \ln 1)$$

$$= \frac{\pi}{4}$$

(A).

5, iii

Date: / /

$$\textcircled{i} \int_0^1 x^3 \sqrt{1+3x^4} \, dx$$

$$\text{let } 1+3x^4 = z^2$$

differentiate both side w.r.t x

$$\int_1^2 \frac{1}{6} z \, dz$$

$$0 + 3 \cdot 4 x^3 \, dx = 2z \, dz$$

$$\Rightarrow \frac{1}{6} \int_1^2 z \, dz$$

$$+ 2x^3$$

$$12x^3 \, dx = 2z \, dz$$

$$\Rightarrow \frac{1}{6} \left[\frac{z^2}{2} \right]_1^2$$

$$\int x^n \, dx = \frac{x^{n+1}}{n+1}$$

$$x^3 \, dx = \frac{z}{12} \, dz$$

$$= \frac{1}{6} \left[\frac{z^2}{2} - \frac{1}{2} \right]$$

x	0	1
z	1	2

$$= \frac{1}{6} \times \left[\frac{8}{2} - \frac{1}{2} \right]$$

$$= \frac{1}{6} \times \frac{7}{2}$$

$$= \frac{7}{12} \text{ (Ans)}$$

$$5-10-(i) \int \left(\frac{x + \sin x}{1 + \cos x} \right) dx$$

$$= \int \frac{x}{1 + \cos x} + \frac{\sin x}{1 + \cos x} dx$$

$$= \int \left(\frac{x}{2 \cos^2 \frac{x}{2}} + \frac{2 \sin \frac{x}{2} \cos \frac{x}{2}}{2 \cos^2 \frac{x}{2}} \right) dx$$

~~$$= \frac{1}{2} \int \frac{x}{\cos^2 \frac{x}{2}} dx$$~~

$$= \frac{1}{2} \int x \sec^2 \frac{x}{2} + \int \tan \frac{x}{2} dx$$

$$= \frac{1}{2} \left[x \cdot \frac{\tan \frac{x}{2}}{\frac{1}{2}} - \int 1 \cdot \tan \frac{x}{2} \times 2 \right] + \int \tan \frac{x}{2} dx$$

$$= x \tan \frac{x}{2} - \int \tan \frac{x}{2} dx + \int \tan \frac{x}{2} dx$$

$$= x \tan \frac{x}{2} + c$$

Ans

5(b) - ii

$$\int \frac{dx}{(\sin x + \cos x)^2}$$

$$= \int \frac{dx}{\left(\frac{\tan x}{\sec x} + \frac{1}{\sec x}\right)^2}$$

$$= \int \frac{dx}{\frac{(1 + \tan x)^2}{\sec x}}$$

$$= \int \frac{\sec x}{(1 + \tan x)^2} dx$$

$$= \int \frac{dt}{(1+t)^2}$$

$$\sin x = \frac{\tan x}{\sec x}$$

$$\cos x = \frac{1}{\sec x}$$

Putting

$$\tan x = t$$

$$\therefore \sec x dx = dt$$

Putting $1+t = u$

$$dt = du$$

$$= \int \frac{du}{u}$$

$$= -\frac{1}{u} + C$$

$$= \frac{-1}{(1+t)} + C$$

$$= \frac{-1}{(\tan x + 1)} + C$$

B(6) iii

$$\int \frac{1}{a+b \sin x} dx$$

$$\sin x = \frac{2 \tan \frac{x}{2}}{1 + \tan^2 \frac{x}{2}}$$

$$= \int \frac{1}{a+b \left(\frac{2 \tan \frac{x}{2}}{1 + \tan^2 \frac{x}{2}} \right)} dx$$

$$= \int \frac{1 + \tan^2 \frac{x}{2} dx}{a + a \tan^2 \frac{x}{2} + 2b \tan \frac{x}{2}}$$

$$= \int \frac{\sec^2 \frac{x}{2} dx}{a \tan^2 \frac{x}{2} + 2b \tan \frac{x}{2} + a}$$

Putting

$$\tan \frac{x}{2} = t$$

$$\sec^2 \frac{x}{2} \cdot \frac{1}{2} dx = dt$$

$$\sec^2 \frac{x}{2} dx = 2dt$$

Now

$$= \int \frac{2dt}{at^2 + 2bt + d}$$

$$= \frac{2}{a} \int \frac{dt}{t^2 + \frac{2b}{a}t + 1}$$

$$= \frac{2}{a} \int \frac{dt}{\left(t + \frac{b}{a}\right)^2 + \frac{a^2 - b^2}{a^2}}$$

$$= \frac{2}{a} \int \frac{dt}{\left(t + \frac{b}{a}\right)^2 + \left(\frac{\sqrt{a^2 - b^2}}{a}\right)^2}$$

$$= \frac{2}{a} \cdot \frac{1}{\frac{\sqrt{a^2 - b^2}}{a}} \cdot \tan^{-1} \left(\frac{t + b/a}{\sqrt{a^2 - b^2}/a} \right) + C$$

$$t^2 + \frac{2b}{a}t + \frac{b^2}{a^2} - \frac{b^2}{a^2} + 1$$

$$\Rightarrow \left(t + \frac{b}{a}\right)^2 + \frac{a^2 - b^2}{a^2}$$

$$\int \frac{1}{x^2 + a^2} = \frac{1}{a} \tan^{-1} \frac{x}{a}$$

$$= \frac{2}{\sqrt{a^2 - b^2}} \tan^{-1} \left(\frac{at + b}{\sqrt{a^2 - b^2}} \right) + C$$

$$= \frac{2}{\sqrt{a^2 - b^2}} \tan^{-1} \frac{a \tan \frac{x}{2} + b}{\sqrt{a^2 - b^2}} + C$$

5(c)

$$\lim_{x \rightarrow 0} \frac{x - \sin x}{x^3}$$

By using L Hospital Rule.

$$\Rightarrow \lim_{x \rightarrow 0} \frac{1 - \cos x}{3x^2}$$

Again using L-Hospital Rule.

$$\Rightarrow \lim_{x \rightarrow 0} \frac{\sin x}{6x}$$

$$\Rightarrow \frac{1}{6} \cdot \lim_{x \rightarrow 0} \frac{\sin x}{x}$$

$$\Rightarrow \frac{1}{6} \times 1$$

$$= \frac{1}{6} \quad \text{proved}$$